

Bayesian Learning

Lecture 12 - Course summary

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Overview

- Wait, what did we actually do?

- **Subjective probability**
- **Bayesian Learning:** Bayes theorem

$$p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{p(\mathbf{x})} \propto p(\mathbf{x}|\theta)p(\theta)$$

- The proportional constant is actually the **marginal likelihood**

$$p(\mathbf{x}) = \int p(\mathbf{x}|\theta)p(\theta)d\theta$$

used for model comparison.

Simple models - conjugate priors

- iid **Bernoulli model** - Beta prior
- iid **Poisson model** - Gamma prior
- iid **Exponential model** - Gamma prior
- iid **Normal known variance** - Normal prior

Multi-parameter models

■ Marginalization

$$p(\theta_1|\mathbf{x}) = \int p(\theta_1, \theta_2|\mathbf{x}) d\theta_2$$

- iid **normal with unknown variance** $N(\theta, \sigma^2)$. Marginal posterior for θ is student- t

$$\theta|\mathbf{x} \sim t\left(\mu_n, \frac{\sigma_n^2}{\kappa_n}, \nu_n\right)$$

- **Multinomial model** - Dirichlet prior
- **Dirichlet distribution** on unit simplex $\sum_{k=1}^K \theta_k = 1$.

■ Different types of priors:

- ▶ **Expert** elicitation
- ▶ **Other data** sources
- ▶ **Non-informative priors** (prior sample size). Bernoulli model:

$$\theta | \mathbf{x} \sim \text{Beta}(\alpha + s, \beta + f)$$

- ▶ **Jeffreys' prior** $p(\theta) = |I(\theta)|^{1/2}$
- ▶ **Hierarchical priors**. L2-regularization

$$p(\beta, \sigma^2, \lambda) = p(\beta | \sigma^2 \lambda) p(\sigma^2) p(\lambda)$$

- ▶ **Regularization** and **smoothness priors**

More later today!